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Bangladesh's energy crisis exposed the cost of overreliance on imported fossil fuels. With viable wind resources along its coast, the country's challenge is no longer feasibility but system design – integrating transmission, hybrid generation and storage to turn coastal wind into reliable, cost-effective national supply

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The country's first wind power plant is situated at Khurushkul in Cox's Bazar, the wind-based plant began commercial production on 8 March 2024. Photo: Syed Zakir Hossain

While Bangladesh welcomes the newly elected government, the nation waits to see how prepared it is for the challenges ahead. Politics may dominate headlines, but ultimately it is policy that determines public satisfaction. Over the past 17 years, governance operated with limited parliamentary challenge. This time, however, the assembly will scrutinise every major decision.

This generation possesses both the knowledge and the tools to evaluate whether new policies are genuinely transformative or merely recycled approaches presented in new language. Encouragingly, the government has access to the same information and analytical resources. How effectively it identifies past mistakes, confronts structural weaknesses, and designs context-specific solutions will ultimately shape its political fate.

When policymakers examine which sector placed the greatest strain on national reserves, corruption may be the easiest answer. Beyond that, however, the power and energy sector stands out. The previous administration's structural overreliance on volatile imported fossil fuels was a primary catalyst of the economic crisis. By neglecting domestic resource development and delaying adaptation to renewable alternatives, Bangladesh remained tethered to expensive imports that rapidly depleted foreign reserves.

Development funds were diverted to plug short-term supply gaps – temporary fixes that burned through national wealth without building resilient infrastructure. The failure to modernise the grid and invest in storage further exposed the system to global market shocks. The result is the vulnerability the country confronts today.

The policy implication is clear: Bangladesh must transition towards sustainable and cost-effective electricity generation anchored in renewable energy. Critics often argue that renewables are too costly or unreliable for Bangladesh. Regional evidence challenges this narrative.

In 2022, renewables accounted for approximately 22% of total energy supply in India and Pakistan, 70% in Nepal, and

81% in Bhutan, according to IRENA (2025). Each country relies on the renewable resource best suited to its geography. In 2024, 48% of India's renewable electricity came from solar and 23% from hydro. Pakistan sourced 66% of its renewables from hydro, 21% from solar, and 11% from wind. Nepal derived 95% of its renewable energy from hydro, while Bhutan's renewable portfolio is almost entirely hydro-based.

The lesson is not that Bangladesh should replicate its neighbours, but that each country has leveraged its geographic advantages. Bangladesh must do the same.

Critics of solar cite land scarcity. Critics of wind point to variable and low-speed conditions. Yet wind variability exists everywhere. The real question is whether the system is designed to manage that variability without burdening consumers.

Wind speeds along the Bay of Bengal range between 5.75 and 7.75 metres per second, classified as significant yet largely underutilised, according to SANEM (2024). Despite this, Bangladesh has only 62MW of installed wind capacity, primarily from one major project, according to SREDA (2026).

Wind has demonstrated proof of performance. The 55MW Mongla wind project has shown both technological and commercial viability, generating electricity at approximately Tk12.5 per kilowatt-hour. In contrast, LNG-based generation cost Tk41.50 per kilowatt-hour in 2022, according to SANEM (2024).

If wind is competitive, why has expansion stalled? The constraint is not wind itself, but the surrounding system.

First, geography and transmission. Wind resources are concentrated along the coast, while demand is centred in Dhaka and other urban hubs. Without robust transmission infrastructure, coastal generation cannot become dependable national supply. The bottleneck is the grid.

Second, capital intensity and technological dependence. Heavy reliance on imported turbines and foreign expertise raises costs and limits domestic capability building.

Third, public confidence. Coastal communities, accustomed to infrastructure failure during severe weather, are understandably sceptical. Trust is built through reliable performance during crises, not project announcements.

Finally, inadequate storage and grid upgrades limit the integration of variable renewables. Intermittency becomes disruptive only when connected to a system without adequate buffering capacity. It is not a fatal flaw; it becomes problematic when variable generation feeds into an inflexible system.

A practical response rests on three pillars: appropriate technology, integrated system design, and shared infrastructure.

First, turbine selection must reflect Bangladesh's wind conditions. Conventional horizontal-axis turbines may underperform in low and variable wind environments. Vertical-axis wind turbines may operate more effectively in multi-directional and lower-speed conditions, particularly in coastal areas of the Bay of Bengal. In cyclone-prone regions, their design characteristics may also offer resilience and maintenance advantages.

Second, wind should not be treated as a standalone investment when hybrid systems can smooth supply. Coastal locations including Mongla, Sandwip and Saint Martin's Island have been identified as suitable for wind-solar hybrid configurations. Floating hybrid systems could further address land constraints.

The logic is straightforward. If one source dips, the other can support generation. During low-pressure systems when cloud cover reduces solar output, higher wind speeds can support electricity production. Variability does not disappear, but the risk of sudden supply shocks declines.

Third, storage must be treated as national infrastructure. Government-operated centralised storage at substations could absorb surplus generation and release it during deficits, enhancing reliability, particularly in weather-vulnerable coastal zones.

Shared transmission lines, substations and storage facilities would reduce duplication and lower overall infrastructure costs. In a capital-constrained country, consolidation is prudent.

There is also a clear investment signal. With government-backed storage lowering investor risk, renewable electricity generation costs under enabling conditions have been projected to potentially fall to Tk3–Tk4 per unit. While timelines may vary, the policy signal is evident: strategic public investment can crowd in private capital by reducing uncertainty and private investment costs.

The real constraint is not nature. It is system design. Bangladesh does not lack wind, nor does it lack ingenuity. What it has lacked is a power system built around reliability rather than reaction.

Coastal communities have long endured fragile supply, and severe weather events repeatedly expose the cost of postponing structural reform. The moment demands more than incremental capacity expansion. It requires deliberate redesign of the power architecture – strengthening transmission before scaling generation, treating storage as core infrastructure rather than a luxury, and aligning technology choices with local wind realities.

If coastal wind is integrated with discipline and foresight, it can become more than an experiment; it can become a pillar of economic stability. The next phase of Bangladesh's energy transition must move beyond debating feasibility and begin engineering certainty. Reliability should not be promised. It should be built into the system from the outset.

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